

Indian Institute of Technology Bombay

PH 544 - Mid Semester Exam General relativity Date: 25/02/2026
 Marks 25 = Weightage 25 points LC-002 Time: 17:00-19:00

1. A curve is specified by the following coordinates in terms of a parameter t :

$$x^0 = \sqrt{3}ct \quad x^1 = ct \quad x^2 = ct_0 \cos\left(\frac{t}{t_0}\right) \quad x^3 = ct_0 \sin\left(\frac{t}{t_0}\right).$$

Determine whether the tangent vector at a typical point on the curve is space-like or time-like assuming that the space-time is Minkowskian. [2]

2. Inertial frame S observes event 1 located at $x_1 = 0\text{m}$ at an instant $t_1 = 0\text{s}$ and observes event 2 at $x_2 = 36 \times 10^5 \text{km}$ with $t_2 = 20\text{s}$. Is it possible to find an inertial frame S' in which the above two events are observed at the same location? If yes, compute the relative velocity of frame S' with respect to S ? If no, justify. [2]

3. Compute the covariant derivative of g_{ij} . [2]

4. Two inertial coordinate systems S and S' move with speed $c/2$ with respect to each other. Draw the space-time diagram relating these two. Let ct and x axis for S be at right angles in your drawing.

(a) Draw calibration hyperbolas that allow you to define unit distances along x', ct' , x and ct axes. [1]

(b) Plot the following points: (1) $x=1, ct=1$ (2) $x'=1, ct'=1$ [2]

5. Compute the Gaussian curvature of the 2-D surface given by $z = f(x, y)$. Here f is a smooth real function. [3]

6. Consider the following fourth-rank tensor in 4-D space-time:

$$T_{\mu\nu\sigma\delta} = \lambda (g_{\mu\sigma} g_{\nu\delta} - g_{\mu\delta} g_{\nu\sigma}) \quad \text{where } \lambda \text{ is a constant}$$

(a) Does this tensor satisfy all the properties of Riemann tensor? Prove Yes/No. [2]

(b) If yes, obtain Ricci tensor, Ricci scalar. (If No, construct a non-zero second rank tensor and scalar from this tensor.) [2]

7. In the class, we derived the Newtonian deviation equation:

$$\frac{d^2 \xi^i}{dt^2} = -K_{ij} \xi^j \quad \text{where } K_{ij} = \frac{\partial^2 \Phi(x^k)}{\partial x^i \partial x^j}$$

here ξ^i is the spatial separation between two near-by particles in Cartesian coordinates, $\Phi(x^k)$ is the Newtonian gravitational potential and K_{ij} is computed at the location of the first particle.

- (a) Calculate the matrix (K_{ij}) due to Moon's gravitational force on the Earth. [2.5]
- (b) Consider the case when the first particle is located at $x^i = (0, 0, r)$. Write down the Newtonian deviation equation for the displacement of the second particle. [1.5]

8. **Rindler metric:** Consider the following 2-dimensional metric:

$$ds^2 = -dT^2 + dX^2$$

where T and X are the two coordinates. Physically, T refers to time and X refers to the space coordinate.

- (a) Show that the following transformation: [2]

$$T = x \sinh(gt) \quad X = x \cosh(gt) \quad \text{where } g \text{ is a constant}$$

leads to the following line-element

$$ds^2 = -g^2 x^2 dt^2 + dx^2$$

- (b) Provide the range of values x and t can take. [1]
- (c) Compute all the non-zero Christoffel symbols for the above coordinate system. [2]

All the best